

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0474

Talbert Tap 138 kV
April 18, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0474. The Interconnection Customer proposes a 210.4 MW solar photovoltaic facility interconnecting at the Talbert Tap 138 kV substation in the Maryland area. The request was received on May 06, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 197 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$663,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rush-Campus Rive 138kV Line	Glenannan Substation-Taber 345kV	161.1	234	5.3
Jerome-Sandyspring 138kV	Glenannan-Taber 345kV	157.2	244	5.1
Campus Drive-Bladensburg 138kV	Taber-Glenannan Substation 345kV Line	139.3	332	5.3
Prairie Wind Farm-Montpelier 138kV Line	Taber-Glenannan 345kV	128.4	397	5.5
Sunny Hills-Bladensburg 138kV	Helm'S Pumped Storage-Glenannan Substation 138kV Line	113.3	586	5.6
Mcewen-Sunny Hills 138kV	Helm'S Pumped Storage-Glenannan 138kV	111.9	619	5.6
Port Robson Substation-Tazeell Substation 138kV	Dixleta-Taber 345kV	103.3	438	5.1
Taber-Tazewell 138kV	Dixleta-Taber 345kV L	100.8	520	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rush-Campus Drive Substation 138kV	Rebuild line with higher-capacity conductor	\$1,921,000	6% (\$108,000)
Jerome-Sandy Spring 138kV	Rebuild line with higher-capacity conductor	\$1,408,000	6% (\$79,000)
Campus Drive-Bladensburg 138kV	Rebuild line with higher-capacity conductor	\$3,118,000	6% (\$175,000)
Prairie Wind Farm-Montpelier 138kV	Rebuild line with higher-capacity conductor	\$2,271,000	6% (\$141,000)
Sunny Hills-Bladensburg 138kV	Rebuild line with higher-capacity conductor	\$1,415,000	2% (\$29,000)
Sunny Hlls-Mcewen 138kV	Rebuild line with higher-capacity conductor	\$828,000	2% (\$17,000)
Port Robson-Tazewell 138kV	Reconductor line	\$499,000	8% (\$39,000)
Tazewell-Taber 138kV	Reconductor line	\$970,000	8% (\$75,000)
	Total allocated to project		\$663,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.